Implement clock gating on a simple counter and analyse the power reduction during synthesis and layout using a low power synthesis flow

1. PROBLEM STATEMENT:

Power dissipation is one of the key issues in contemporary VLSI technology. In digital designs, the clock signal is always toggling and is used to drive all the sequential components like flip-flops. Even during idle periods, the clock signal is always switching, resulting in unnecessary dynamic power dissipation.

The key contributors to power dissipation in digital designs are:

- Clock network (30-50% of total power)
- Flip-flops
- Combinational logic

Total power dissipation in digital designs is expressed as:

P(total)=P(dynamic)+P(short−circuit)+P(static)

Dynamic power is primarily due to:

- Switching activity
- Charging and discharging of capacitances
- Clock frequency
- Supply voltage

Since dynamic power is very sensitive to switching activity, minimizing unnecessary clock switching is an effective means of reducing power dissipation.

Objective of the project:

To apply clock gating to a simple counter and measure the power savings during synthesis and layout stages using a low-power synthesis flow.

2. IMPLEMENTATION:

Concept Applied: Clock Gating

Clock gating is a low-power design methodology applied to minimize dynamic power dissipation by turning off the clock when the circuit activity is not needed.

Instead of allowing the clock to switch continuously, the clock is managed by an enable signal.

Enable = 0 → Clock is turned off → No switching activity

Enable = 1 → Clock is enabled → Counter works as usual

This minimizes:

1. Unnecessary clock switching
2. Dynamic power dissipation

Integrated Clock Gating (ICG) Logic:

In this project, Integrated Clock Gating (ICG) is applied.

ICG comprises:

- Clock input
- Enable signal
- Latch
- AND gate logic

Operation:

- ➢ The latch eliminates glitches on the gated clock.
- ➢ The AND gate manages the clock to reach the counter.
    - ▪ Enable = 0 → Clock is turned off safely.
    - ▪ Enable = 1 → Clock reaches the counter.

Benefits:

- Glitch-free clock switching
- Less switching activity
- Less dynamic power dissipation
- Applicable for backend low-power synthesis and physical design

Counter Using Clock Gating:

A simple counter is implemented using T flip-flops.

The clock is gated using the enable signal before it reaches the flip-flops.

The T flip-flops are combined in multiple stages to make the counter.

Operation:

If Enable is LOW → The gated clock is OFF → The flip-flops don't toggle.

If Enable is HIGH → The gated clock is ON → The counter increments.

3. VERILOG CODE:

Counter Using Clock Gating :

```
module clock_gate ( input  clk, input  en,  input  clr, output gclk);
 reg en_latch;
  // Latch enable when clock is low (safe clock gating)
always @(negedge clk or posedge clr) begin
        if (clr)
            en_latch <= 1'b0;
        else
            en_latch <= en;
    end
// Gated clock
    assign gclk = clk & en_latch;
endmodule

module counter(clk,en,clr,q);
input clk,en,clr;
output [3:0]q;
wire gclk;
wire t0,t1,t2,t3;
clock_gate cg1(clk,en,clr,gclk);
```

```
assign t0=en;
assign t1=en&q[0];
assign t2=en&q[0]&q[1];
assign t3=en&q[0]&q[1]&q[2];
tff tf1(t0,gclk,clr,q[0]);
tff tf2(t1,gclk,clr,q[1]);
tff tf3(t2,gclk,clr,q[2]);
tff tf4(t3,gclk,clr,q[3]);
endmodule

module tff(t,clk,clr,q);
input clk,clr,t;
output q;
reg q;
always@(negedge clk or posedge clr)
begin
if(clr==1)
 q<=0;
else
 case(t)
  1'b0:q<=q;
1'b1:q<=~q;
endcase
end
endmodule

module testbench();
reg clk,en,clr;
wire [3:0]q;
counter aa(clk,en,clr,q);
initial
begin
```

```
clk=0;clr=1;en=0;
#1 clr=0;
#10 en=1;
#160 en=0;
end
always #5clk=~clk;
endmodule
```

4. OUTPUT:

Fig. 1: RTL Schematic of Clock Gated Counter

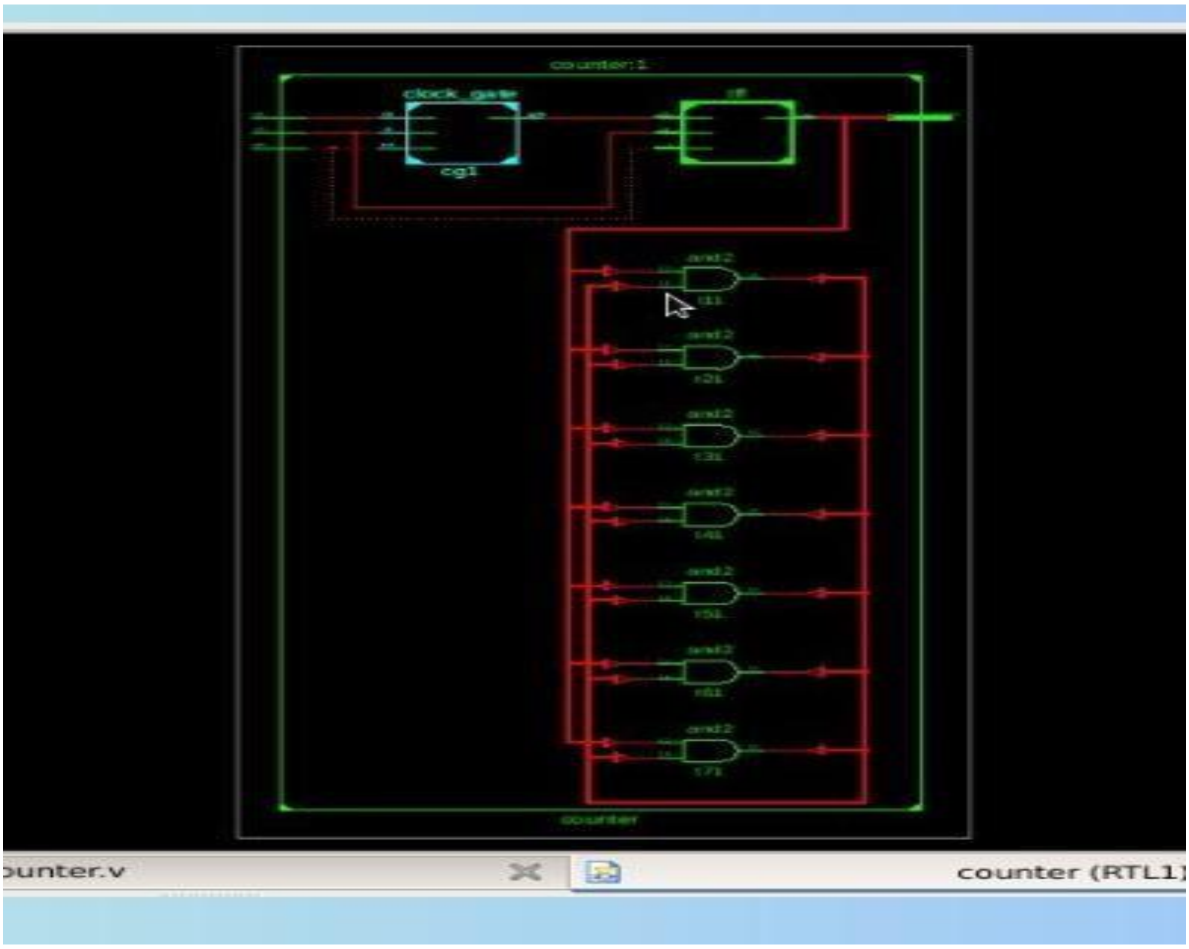

Fig.2: Simulated Output of the Clock Gated Counter

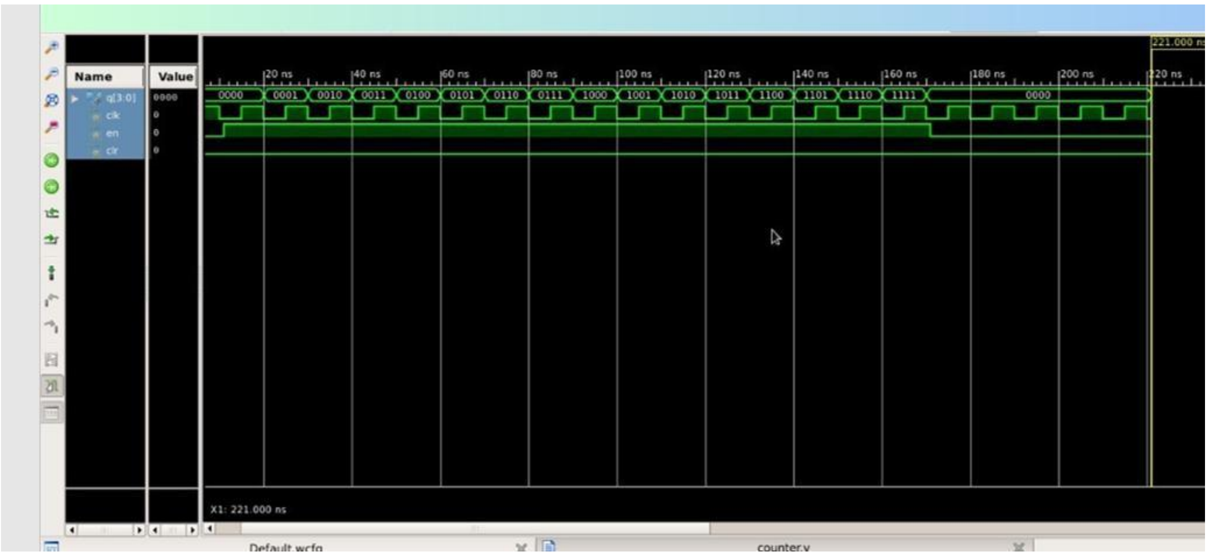

Fig.3: Power Report without Clock Gating :

```
Instance: /clock
Power Unit: W
PDB Frames: /stim#0/frame#0
  -----------------------------------------------------------------------
--
    Category         Leakage     Internal    Switching        Total
Row%
  -----------------------------------------------------------------------
--
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
    register     9.38370e-09  9.02823e-06  5.09134e-07  9.54675e-06
97.49%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
       logic     9.10984e-10  1.44919e-07  9.97272e-08  2.45557e-07
2.51%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
       clock     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
  -----------------------------------------------------------------------
--
    Subtotal     1.02947e-08  9.17315e-06  6.08861e-07  9.79231e-06
100.00%
  Percentage           0.11%       93.68%        6.22%      100.00%
100.00%
  -----------------------------------------------------------------------
--
```

Power Report with Clock Gating :

Fig.4: Power Report with Clock Gating :

```
Instance: /counter
Power Unit: W
PDB Frames: /stim#0/frame#0
  -------------------------------------------------------------------------
--
    Category         Leakage     Internal    Switching        Total
Row%
  -------------------------------------------------------------------------
--
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
    register     3.14668e-07  9.29219e-07  3.40200e-08  1.27791e-06
88.25%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
       logic     4.71533e-08  8.07879e-08  4.22212e-08  1.70162e-07
11.75%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
       clock     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00
0.00%
  -------------------------------------------------------------------------
--
    Subtotal     3.61821e-07  1.01001e-06  7.62412e-08  1.44807e-06
100.00%
  Percentage          24.99%       69.75%        5.27%      100.00%
100.00%
  -------------------------------------------------------------------------
--
```

Observation:

- The power analysis indicates that the counter without clock gating consumes higher total power (**9.79231×10^{-6} W**) due to continuous clock activity.
- The clock-gated counter shows a significant reduction in total power to **1.44807×10^{-6} W**, demonstrating the effectiveness of clock control.
- A noticeable decrease in switching power is observed in the gated design, confirming reduced unnecessary flip-flop transitions.
- Internal power remains the dominant component in both designs, but it is substantially lower in the clock-gated implementation.
- The results validate that clock gating improves power efficiency by suppressing idle clock propagation in sequential circuits.

COMPARISON TABLE:

Metric	Without Clock Gating	With Clock Gating
Total Power	**9.79231×10^{-6} W**	**1.44807×10^{-6} W**
Leakage Power	1.02947×10^{-8} W	3.61821×10^{-7} W
Internal Power	9.17315×10^{-6} W	1.01001×10^{-6} W
Switching Power	6.08861×10^{-7} W	7.62412×10^{-8} W
Clock Activity	Continuous	Enable-controlled
Power Efficiency	Higher power consumption	Reduced power consumption
Area	No extra logic	Slightly increased

5. Conclusion:

The project successfully demonstrated the implementation of clock gating on a synchronous counter using Cadence Genus low-power synthesis flow. The comparison between gated and non-gated designs illustrates the impact of clock gating on switching activity and power consumption.

Clock gating effectively reduces unnecessary clock transitions in sequential elements and is widely used in modern low-power VLSI design. The work provided practical understanding of low-power synthesis techniques, clock control strategies, and power analysis methodology used in industry-standard tools.